

Linear Shaker & Orbital Shaker

User Manual



SK-L180-E Analog Linear Shaker

SK-O180-E Analog Orbital Shaker

SK-L180-S LED Digital Linear Shaker

SK-O180-S LED Digital Orbital Shaker

Please read carefully before use, and follow operating and safety instructions!

The design and specifications are subject change without prior notice.

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Preface

Welcome to the “Linear Shaker & Orbital Shaker User Manual”. Users should read this manual carefully, follow the instructions and procedures, and beware of all the cautions when using this instrument.

Service

When help needed, you can always contact the service department of manufacturer for technical support in the following ways:

DLAB Scientific Instrument Inc.

2311 E. Locust Court, Ontario, CA 91761 United States.

Office: +1-747-230-5179

Fax: +1-909-230-5275

Sales contact: info@dlabsci.com

Service contact: service@dlabsci.com

www.dlabsci.com

Please provide the customer care representative with the following information:

- Serial number (on the rear panel)
- Certification
- Description of problem (i.e., hardware or software)
- Methods and procedures adopted to resolve the problems
- Your contact information

Warranty

This instrument is warranted to be free from defects in materials and workmanship under normal use and service, for a period of 24 months from the date of invoice. The warranty is extended only to the original purchaser. It shall not apply to any product or parts which have been damaged on account of improper installation, improper connections, misuse, accident or abnormal conditions of operation.

For claims under the warranty please contact your local supplier. You may also send the instrument directly to manufacturer, enclosing the invoice copy and by giving reasons for the claim.

1. Safety Instructions



Warning!

- Read the operating instructions carefully before use.
- Ensure that only trained staff works with the instrument.



Protective ground contact !

- Ensure the socket must be grounded (protective ground contact) before use.

- When working wear personal safety guards to avoid the risk of:
 - Splashing liquids
 - Broken glass containers
- Follow the safety instructions, guidelines and accident prevention regulations.
- Do not touch the running parts, moving instrument care not rolling your fingers.
- Set up the instrument in a spacious area on a stable, clean, non-slip, dry and fireproof surface. Do not operate the instrument in explosive atmospheres, with hazardous substances or under water.

- If the instrument does not run smoothly, please decrease the motor speed.
- Firmly secure the accessories and vessels in place to avoid damage or risk.
- Preparation of samples may lead to dangerous flammable. Only process samples that will not react dangerous.
- Use the standard accessories listed in the “accessories” section, and follow the instructions to use accessories to ensure safety. Please switch off the power before assembly of accessories, confirm the instrument and accessories are intact before switch on each time.
- The instrument only be opened by expert, please switch off before use.
- The voltage stated on the nameplate must correspond to the mains voltage.
- Do not cover the instrument during running. Prevent the collision and extrusion to instrument and accessories.
- Keep away from high magnetic field.

2. Proper Use

The instrument is designed for mixing sticky substance in schools, laboratories or factories. It can be installed on a variety of impeller, for different viscosity of the medium. This instrument is not suitable for using in residential areas or other constraints mentioned in Chapter 1.

Do not use the accessories recommended by the manufacturer, or failure to use the instructions, may be caused unsafe situation.

3. Inspection

3.1 Receiving Inspection

Unpack the instrument carefully and check for any damages which may have arisen during transport. If it happens, please contact manufacturer/supplier for technical support.



Note:

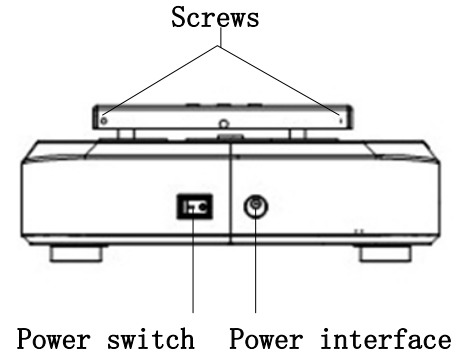
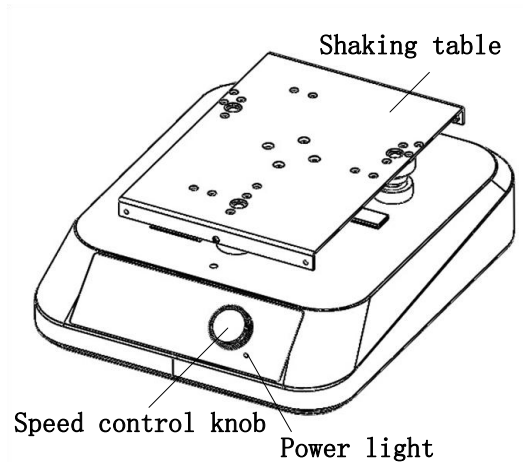
If there is any apparent damage to the system, please do not connect to the power line.

3.2 Listing of Items

Items	Qty
Main unit	1
Power cable	1
User manual	1

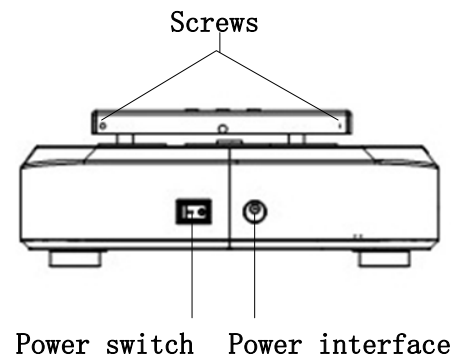
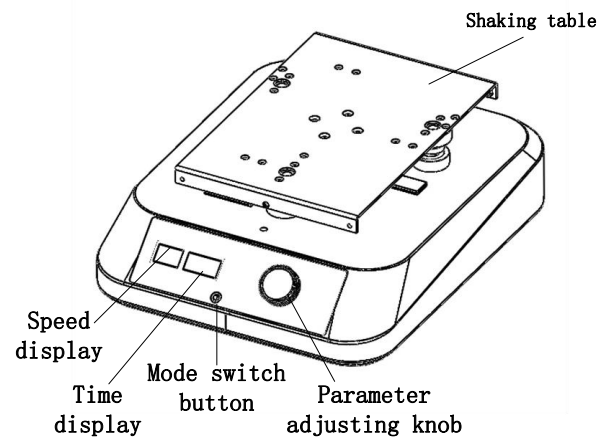
4. Control

4.1 Analog Model



Items	Descriptions
Speed control knob	Set target speed.
Power switch	Switch ON/OFF instrument.
Power interface	Connect the power cable
Power light	Green LED display light
Screw	Install the platform.

4.2 LED Digital Model



Items	Default settings
Mode switch button Mode	Shift speed and time setup mode
Parameter adjusting knob	Set speed and time by rotating the knob. Push on/off to start/stop rotating function.
Speed display	Display the set speed values.
Time display	Display the set time values.
Power switch I/O	Switch ON or OFF the instrument.
Shaking table	Shaking.
Screw	Install the platform.

5. Operation

5.1 Analog Model

- Place the instrument in safe and stable surface, ensure the required operating voltage and power supply voltage matched.
- Connect the power cable.
- Install the platform.
- Switch ON the instrument.
- Rotate the speed control knob clockwise to set target speed, and the shaking function will be switched on.
- Rotate the speed control knob to the left-most position to switch off the shaking function.
- Switch OFF the instrument.

5.2 LED Digital Model

- Place the instrument in safe and stable surface, ensure the required operating voltage and power supply voltage matched.
- Connect the power cable.
- Install the platform.
- Switch ON the instrument.

- The instrument begins self-checking.
- When initialization is over, LED displays the last running values of speed and timer.
- Rotate the speed/timer control knob to set target speed.
 - a. Press the mode switch button to choose the speed or timer setup interface. When the character of LED display is flashing, rotate the parameter adjusting knob to set the target parameter. Press the knob to start the motion.
 - b. During operation, motion can be stopped at any time by pressing the parameter adjusting knob. If the knob is pressed again, motion will start again and the timer will continue to countdown. When the timer reaches zero, the unit will be automatically halted.
 - c. When the timer is set to be 00:00, start the continuous motion.
- Switch OFF the instrument.

If these operations above are normal, the instrument is ready to operate. If not, the instrument may be damaged during transportation, please contact technical support of manufacturer/supplier.

**Note:**

The speed value can be altered during shaking.

6. Faults

- When switched ON, the instrument does not work
 - Check whether the power cable is connected
- Motor speed cannot reach set value
 - Check whether it is overloaded

7. Maintenance and Cleaning

- Proper maintenance can keep instrument working properly and lengthen its lifetime.
- Do not spray cleanser into the instrument when cleaning.
- Do not remove the power line when cleaning. Only use recommended cleansers:

Dyes	Isopropyl alcohol
Construction Materials	Water containing tenside / Isopropyl alcohol
Cosmetics	Water containing tenside /

	Isopropyl alcohol
Foodstuffs	Water containing tenside
Fuels	Water containing tenside

- Before using other method for cleaning or decontamination, the user must ascertain with the manufacturer that this method will not damage the instrument. Wear the proper protective gloves during cleaning of the instrument.

**Note:**

- Electronic device cannot clean with cleanser.
- If you require maintenance service, must be cleaned the instrument in advance to avoid pollution of hazardous substances, and to send back into original packing.
- If the instrument will not use for a long time, please switch off and place in a dry, clean, room temperature and stable location.

8. Associated standards and regulations

Construction in accordance with the following safety standards:

EN 61010-1

UL 3101-1

CAN/CSA C22.2(1010-1)

EN 61010-2-10

Construction in accordance with the following EMC standards:

EN 61326-1

Changes or modifications not expressly approved by the party responsible for compliance could void the user's authority to operate the equipment.

NOTE: This equipment has been tested and found to comply with the limits for a Class A digital device, pursuant to Part 15 of the FCC Rules. These limits are designed to provide reasonable protection against harmful interference when the equipment is operated in a commercial environment. This equipment generates, uses, and can radiate radio frequency energy and, if not installed and used in accordance with the instruction manual, may

cause harmful interference to radio communications. Operation of this equipment in a residential area is likely to cause harmful interference in which case the user will be required to correct the interference at his own expense.

9. Specification

Items	Specifications	
	Analog Model	LED Digital Model
Voltage [VAC]	100-240	100-240
Frequency [Hz]	50/60	50/60
Motor rating input[W]	16	16
Motor rating output[W]	10	10
Shaking motion	Linear/Orbital	Linear/Orbital
Orbital diameter[mm]	20	20
Max. load capacity (with platform) [Kg]	3	3
Motor type	DC motor	DC motor
Speed range[rpm]	0-200	70-200
Dimensions[W×H×D mm]	330 × 270 × 105	330×270×105

Items	Specifications	
	Analog Model	LED Digital Model
Operation type	Continuous	Continuous/timed
Time setting range	-	1min-19h59min
Weight[kg]	3.1	3.1
Permissible ambient temperature[°C]	5-40	5-40
Permissible relative humidity	80%	80%
Protection class acc. DIN EN60529	IP21	IP21

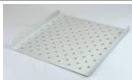
10. Order Information

Cat. No.	Descriptions
832113017777	SK-O180-E, Orbital Analog Shaker, US plug, 100V-240V, 50Hz/60Hz
832113117777	SK-O180-E, Orbital Analog Shaker, CN plug, 100V-240V, 50Hz/60Hz
832113217777	SK-O180-E, Orbital Analog Shaker, Euro plug, 100V-240V, 50Hz/60Hz
832113317777	SK-O180-E, Orbital Analog Shaker, UK plug, 100V-240V, 50Hz/60Hz
831112017777	SK-L180-E, Linear Analog Shaker, US plug, 100V-240V, 50Hz/60Hz
831112117777	SK-L180-E, Linear Analog Shaker, CN plug, 100V-240V, 50Hz/60Hz
831112217777	SK-L180-E, Linear Analog Shaker, Euro plug, 100V-240V, 50Hz/60Hz

831112317777	SK-L180-E, Linear Analog Shaker, UK plug, 100V-240V, 50Hz/60Hz
832111017777	SK-O180-S, LED digital Orbital Shaker, US plug, 100V-240V, 50Hz/60Hz
832111117777	SK-O180-S, LED Digital Orbital Shaker, CN plug, 100V-240V, 50Hz/60Hz
832111217777	SK-O180-S, LED Digital Orbital Shaker, Euro plug, 100V-240V, 50Hz/60Hz
832111317777	SK-O180-S, LED Digital Orbital Shaker, UK plug, 100V-240V, 50Hz/60Hz
831111017777	SK-L180-S, LED Digital Linear Shaker, US plug, 100V-240V, 50Hz/60Hz
831111117777	SK-L180-S, LED Digital Linear Shaker, CN plug, 100V-240V, 50Hz/60Hz

831111217777	SK-L180-S, LED Digital Linear Shaker, Euro plug, 100V-240V, 50Hz/60Hz
831111317777	SK-L180-S, LED Digital Linear Shaker, UK plug, 100V-240V, 50Hz/60Hz

Accessories

	Cat. No.	Descriptions
	18900025	SK 180.1 Universal attachment with 3 bars
	18900026	SK180.2 Fixing clip attachment
	18900038	SK180.3 Lengthways roller attachment
	18900039	SK180.4 Dish Attachment with non-slip mat

	18900029	Fixing clip for flask volume 25 ml
	18900030	Fixing clip for flask volume 50 ml
	18900031	Fixing clip for flask volume 100 ml
	18900032	Fixing clip for flask volume 200/250 ml
	18900033	Fixing clip for flask volume 500 ml
	18900037	Bar for SK180.1 Universal attachment, black



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